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Basic Details of the Team and Problem Statement

Problem Statement ID – SIH26082

Problem Statement Title- Air Pollution–Weather Coupled

Forecasting System (Delhi NCR Focus)

Theme- Disaster Management

PS Category- Software

Team ID-

Team Name (Registered on portal)- atom(x)

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Real-Time Air Quality Explorer with Maps & AI

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❖ Unique Idea/Solution:

Our solution leverages atmospheric physics-coupled forecasting, structured databases, and interactive dashboards to enable natural language querying, exploration, and visualization of coupled air quality and meteorology data, democratizing access for researchers, policymakers, students, and non-technical users.

Conversational Access

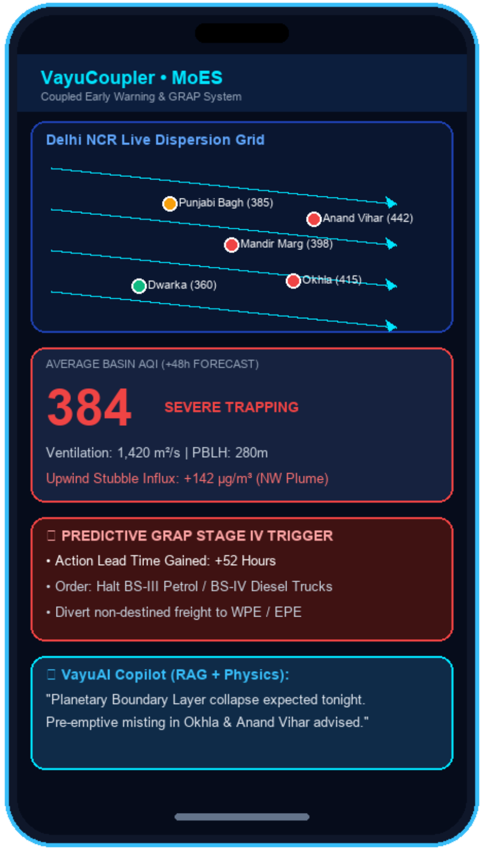
First-of-its-kind chatbot interface for Delhi NCR air quality, allowing users to ask questions in natural language (no coding/WRF expertise needed) on Past Data & Latest Data.

Integrated Geospatial Maps

Users can explore 16 Delhi NCR monitoring stations on an interactive map and directly query data, animated wind streamlines, and stubble fire clusters from specific regions.

Hybrid Query & Physics System

Combines structured sensor feeds with atmospheric physics equations (Ventilation Index, Inversion Factor K_trap) and Fine-Tuned LLM-driven interpretation using RAG + MCP.



Real Time Air Quality Exploration using Interactive Maps.

Conversational Query System using LLM + RAG Pipelines

Automatic Ingestion of CPCB Sensors + NASA FIRMS Satellites

Atmospheric Boundary Layer (PBLH) & Inversion Coupling

Geospatial Tracking of Transboundary Stubble Fire Plumes & Trajectories

Delivers direct scientific insights for public health, municipal bodies & policy.

❖ **OUR PROTOTYPE UI - LINK**

❖ **OUR Repository - LINK**

❖ **Prototype Status:**
100% Prototype completed; live multi-platform suite (Vercel Web App, Windows Offline App, and Android APK built).

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TECHNICAL APPROACH

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using Interactive Maps to Enable region-based search and intuitive data discovery.

Converts multi-source feeds into structured databases, simplifying access and enabling fast, queryable insights.

Answering Queries on Features like PM2.5, PM10, Ventilation Index (VI), PBLH, Thermal Inversion (ΔT)

Using Figma To Designing Our website
Using React.Js / HTML5 for building interface of website

1. HETEROGENEOUS DATA INGESTION

- 16 CPCB Ground Monitoring Stations (PM2.5, PM10, NO_2, CO, SO_2)
- IMD High-Altitude Radiosonde Soundings (PBLH, Wind, Inversion ΔT)
- NASA FIRMS Satellite Active Fire Hotspots (Punjab & Haryana)

2. ATMOSPHERIC PHYSICS & COUPLING ENGINE

- Ventilation Index Modulator: VI = Wind Speed × PBLH (m^2/s)
- Nocturnal Thermal Inversion Trapping Factor: K_trap
- Upwind Stubble Transport Vector Dot Product: S_vector

3. COUPLED FORECASTER & DECISION ENGINE

- +24h, +48h, +72h Predictive AQI Forecast with 90% Confidence Bands
- Automated Predictive GRAP Rules Engine (Stages I to IV Pre-Emptive)
- Counterfactual 'What-If' Simulator (Stubble & Truck Ban Impact)

4. MULTI-AGENCY ACTION DISPATCH LAYER

- Punjab & Haryana Agri: Bio-decomposer & Happy Seeder triggers
- Traffic Police: Inter-state commercial truck bypass to WPE/EPE
- Municipal Corporations (MCD): Targeted smog guns & anti-dust misting

5. UNIFIED DELIVERY COMMAND CENTERS

- MoES Command Center Dashboard (Live Production on Vercel)
- 100% Offline Windows Desktop App (Zero-Install HTML5/PWA)
- Standalone Android Mobile APK (Field Officer Direct Dispatch)

using LLM + RAG Pipelines to Allow non-technical users to ask questions in plain language.

Geospatial Visualization of Stubble Plumes & Trajectories → Provides clear insights into smoke transport & coverage.

Using Cloud Deployment Deploying on Vercel & Render for scalability, reliability + 100% offline standalone mode.

Using Node.Js / FastAPI for managing requests, database, and running physics models & GRAP dispatch rules.

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FEASIBILITY AND VIABILITY

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❖ Feasibility:

This project is technically feasible as it leverages existing AI/ML, atmospheric dispersion physics, and verified open geospatial datasets. CPCB ground telemetry, IMD meteorological soundings, and NASA FIRMS satellite feeds are publicly accessible, and with structured conversion plus cloud deployment, the system can scale to handle large volumes of environmental data. The integration of LLMs with geospatial dashboards and physics equations ensures both usability and accuracy, making the solution practical for researchers, policymakers, and educators.

❖ Potential challenges and risks:

➤ **Query Accuracy:** LLM may misinterpret complex atmospheric queries → **Mitigation:** Fine-tuning + RAG with domain-specific meteorological metadata.

➤ **Data Quality & Coverage:** Heavy cloud cover or limited sensors in certain sub-districts → **Mitigation:** Data augmentation + integration with satellite AOD & neighboring stations.

➤ **Computational Demand:** Real-time spatial-temporal dispersion maps require high resources → **Mitigation:** Scalable cloud deployment + vectorized NumPy/Scipy physics modules.

➤ **Multi-Agency Latency:** Bureaucratic delay between forecast and enforcement → **Mitigation:** Automated role-specific pre-drafted dispatch work orders for CAQM & MCD.

❖ Strategies for Overcoming These Challenges:

➤ **RAG + Fine-Tuning:** Improve query accuracy with domain-specific atmospheric embeddings, prompt guardrails, and physics transfer learning.

➤ **Data Optimization:** Use Parquet, compact JSON, and compressed indexed local storage for handling large spatial-temporal datasets at sub-second speed.

➤ **Efficient Visualization:** Use optimized libraries (Leaflet, Canvas, SVG overlays) for smooth 60fps real-time map rendering.

➤ **Multi-Source Integration:** Combine CPCB ground monitors with IMD boundary layer meteorology, NASA FIRMS fire satellites, and wind streamlines to fill coverage gaps.

IMPACT AND BENEFITS

❖ Potential Impact on the Target Audience:

The solution makes air pollution data more accessible for researchers, provides policymakers with reliable predictive insights for planning, supports educators and students in learning, helps municipal & health sectors with proactive pre-emptive monitoring, and aids environmental governance groups in coordinated crisis mitigation.

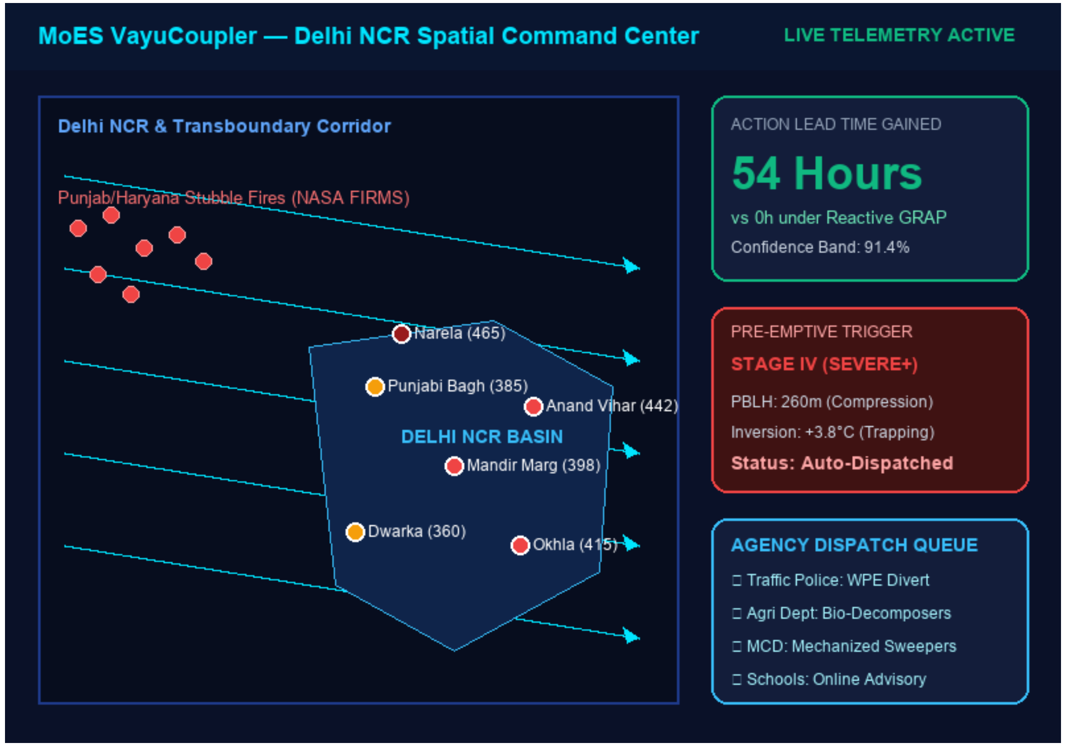
❖ Benefits of the solution:

➤ **Social:** Democratizes access to air quality data for non-technical users, Enhances public awareness, and protects 30M+ citizens via 48h advance health advisories.

➤ **Economic:** Prevents chaotic sudden shutdowns of construction & freight logistics, Enables planned bypasses via EPE/WPE, and Reduces institutional costs of emergency data analysis.

➤ **Environmental:** Improves monitoring of atmospheric boundary layer health and smog indicators, Maximizes anti-smog gun impact before dispersion collapses, and Facilitates early detection of transboundary smoke incursions.

Air Pollution Observation Network



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RESEARCH AND REFERENCES

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1. **SAFAR: Air Quality and Weather Forecasting And Research** Authors: Gufran Beig, S. Parkhi, et al. (2015) **Summary:** Introduces SAFAR-Delhi, the foundational operational framework for coupling meteorological boundary layer dynamics with Eulerian chemical transport models over the National Capital Region. Evaluates boundary layer ventilation coefficients and transboundary dispersion pathways.

2. **NASA FIRMS: Active Fire Detection & Transboundary Smoke Attribution** Authors: Wilfrid Schroeder, Louis Giglio, et al. (2014) **Summary:** Validates 375m VIIRS and MODIS satellite thermal anomaly detection for biomass burning, agricultural stubble fires, and real-time transboundary plume transport tracking across South Asia. Demonstrates robust spatial correlation with downwind particulate matter spikes.

3. **Graded Response Action Plan (GRAP) Guidelines** Authors: Commission for Air Quality Management in NCR & Adjoining Areas (CAQM, 2023) **Summary:** Official statutory emergency response protocol defining Stage I to Stage IV mitigation actions based on AQI thresholds and meteorological dispersion indices. Provides the baseline rulebook that VayuCoupler transforms from reactive to predictive enforcement.

4. **Atmospheric Boundary Layer Dynamics & Inversion Trapping** Authors: Roland B. Stull (1988/2021) **Summary:** Mathematical formulation of Planetary Boundary Layer Height (PBLH), nocturnal thermal inversion capping, and the Ventilation Index (VI = Wind Speed × PBLH) governing basin trapping. Provides the verified physical equations powering VayuCoupler's deterministic coupling engine.